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Technical proposal (Envelope 1)

Tender NGPA/T/2026/052 · Air Quality and Noise Monitoring Network

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/052, we, Kaltenmoor Messtechnik GmbH, submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Dr. Henrik Brandes
Managing Director, Kaltenmoor Messtechnik GmbH

Date: 8 December 2026
Reference: KM-2026-NGPA-052

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1. Company and references

Kaltenmoor Messtechnik has manufactured air-quality analysers and turnkey monitoring stations since 1994. With 210 staff, we have delivered more than 900 stations in 30 countries. Our regional service team (2 engineers) covers the Gulf from outside the State; civil works for this contract will be carried out by a local subcontractor.

Reference	Scope	Year
Lower Elbe Port Authority	6 stations, data platform	2020
Rhine-Ruhr industrial district	12 stations	2022
Baltic Sea Ferry Terminal	3 stations, 4 noise terminals	2024

2. Technical solution

Air-quality stations. Each station is built and tested in our Oldenburg factory:

- AK-PM 300 monitor for PM10 and PM2.5, certified by TÜV as equivalent to the EN reference method (EN 16450); certificates attached;
- AK-NOx chemiluminescence analyser (EN 14211), AK-SO2 UV-fluorescence analyser (EN 14212) and AK-O3 UV-photometric analyser (EN 14625);
- meteorological sensors for wind speed and direction, temperature and humidity;
- a data logger with 7 days of local storage and automatic back-fill, and a 4G router that switches automatically to a second operator.

Shelters. Air-conditioned steel shelters rated IP54, for ambient temperatures from 0 °C to +50 °C, with a 2-hour UPS.

Noise terminals. AK-Sound 61 Class 1 sound level meters to IEC 61672-1, with an all-weather microphone kit, 1/3-octave spectra and audio recording of exceedances.

Data capture. At least 90 % valid hourly data per pollutant and quarter.

AirBase platform. Automatic and manual validation with an audit trail. The alarm module checks the hourly averages every 15 minutes and sends e-mails to the duty officer. Monthly and annual reports export to Excel and PDF. The public dashboard is available in English and German; an Arabic version can be produced on request. Published data is available through a documented REST API.

3. Security and data

Hosting. AirBase is installed on the Authority's servers in its data centre. Users sign in with their Active Directory accounts through LDAP. Raw and validated data are kept for 10 years.

4. Implementation and timeline

Milestone	Month

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Factory acceptance test in Oldenburg	4
Shipment	5
Installation and commissioning	6–7
30-day trial run and acceptance	8

Training. 5 days at our Oldenburg factory for 4 of the Authority's staff, in English.

5. Support, warranty and SLA

Warranty. 24 months from acceptance on all equipment and software.

Operation and maintenance for 3 years. Automatic zero and span checks every 2 weeks, and a full multipoint calibration every 6 months with gases traceable to an ISO/IEC 17025 laboratory. Records are kept for 5 years. Noise terminals are verified every year.

Response. Remote diagnosis within 4 hours; an engineer on site within 5 working days. Spare analysers are shipped from Oldenburg.

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Annex A. Compliance matrix (Annex 1 of the RFP)

Clause	Requirement	Response	Offered
2.1	Six AQ stations: PM10, PM2.5, NO2, SO2, O3 + meteorology	Comply	§2
2.2	Four noise terminals	Comply	§2
3.1	PM analysers certified equivalent (EN 16450)	Comply	TÜV EN 16450 (§2)
3.2	Gas analysers to EN reference methods	Comply	§2
3.3	Noise Class 1, 1/3-octave, audio recording of exceedances	Comply	AK-Sound 61 (§2)
3.4	Shelter IP55, 0 to +50 °C, UPS ≥ 2 h	Comply	IP54, 2 h UPS (§2)
3.5	7 days local storage, back-fill, backup link	Comply	§2
3.6	≥ 90 % valid hourly data per quarter	Comply	§2
4.1	Authority data centre; 10-year retention	Comply	§3
4.2	Validation with audit trail	Comply	§2
4.3	Alarms within 5 min (SMS, e-mail)	Partially comply	E-mail, 15-minute checks (§2)
4.4	Monthly / annual permit reports (Excel, PDF)	Comply	§2
4.5	Public dashboard AR/EN + REST API	Partially comply	English and German; Arabic on request (§2)
4.6	Active Directory sign-in	Comply	LDAP (§3)
5.1	In operation ≤ 6 months; 30-day trial	Partially comply	Acceptance month 8 (§4)
5.2	Calibration every 3 months, ISO/IEC 17025 traceable	Comply	Checks every 2 weeks, calibration every 6 months (§5)
5.3	48 h on-site response; spares in the State	Partially comply	5 working days (§5)
5.4	24-month warranty; 3-year O&M	Comply	§5
5.5	Training of 6 HSE staff, AR/EN	Comply	Factory training (§4)
6.1	≥ 3 similar networks in 7 years	Comply	§1
6.2	Office or service partner in the State	Partially comply	Regional team, local subcontractor (§1)